

6. MASS AND BALANCE

6.1 EQUIPMENT LIST ENTRIES

System/Equipment Designation	Part No.	Mass (kg)	Arm (mm)	Massmoment (kgmm)
SX-16 Searchlight with IR Filter	105-864866	33.50	730 (476)	24455 (15946)
– Installation Provisions	105-864946	8.72	1900 (1646)	16568 (14353)
– Detachable Items	105-864896	24.78	320 (66)	7930 (1635)
NOTE Above values in parentheses apply to Model S Variants.				

7. SYSTEM DESCRIPTION

The high-intensity (1600-Watt), xenon arc lamp equipped searchlight (Nightsun Searchlight, Model SX-16 with IR IFCO) is mounted on the fuselage nose below the battery compartment. The searchlight and IR filter are remotely controlled from a panel in the center console.

The searchlight assembly comprises a searchlight with IR filter assembly, a gimbal, a junction box, a control panel and a four-way switch on the RH collective pitch lever. The searchlight and gimbal are secured to a quick-removable mount. The junction box is mounted in the cargo compartment on the RH side. The control panel contains the switches necessary for searchlight operation.

The following table describes the switches and their functions:

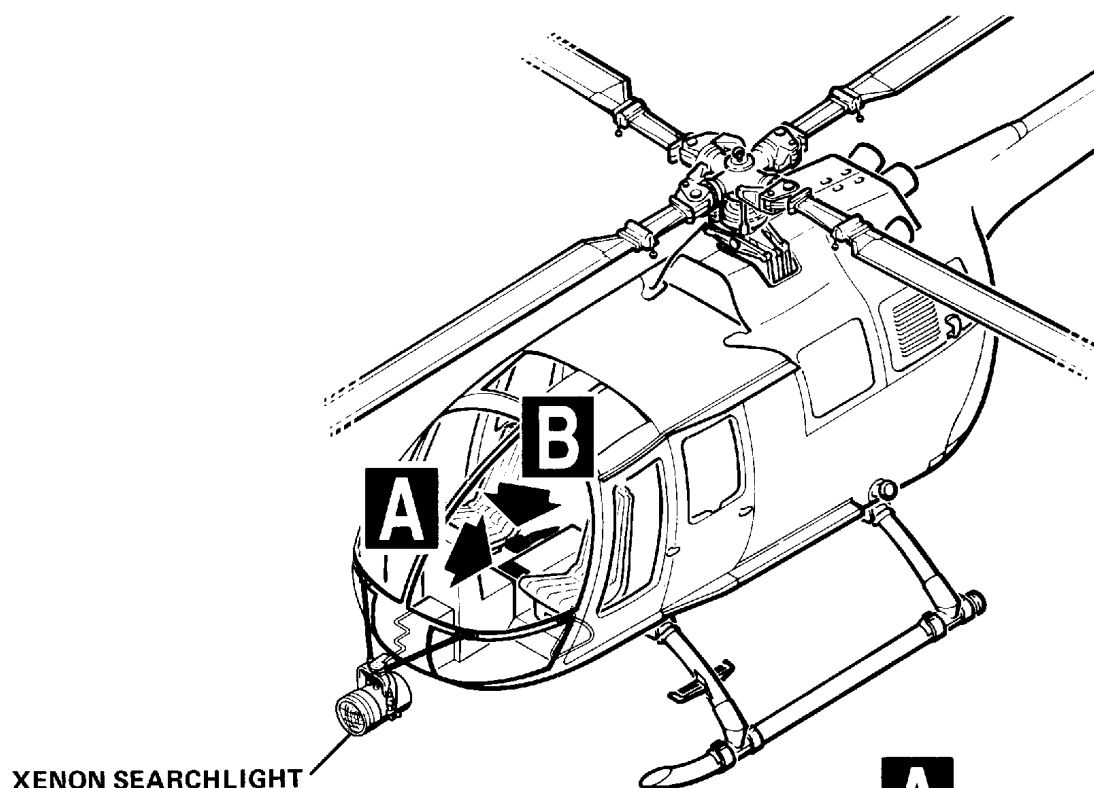
SWITCH	POSITION	FUNCTION
MASTER	ON	Acts as a circuit breaker and supplies power for searchlight lamp (including IR filter) and controls, and integral cooling fan.
	OFF	All searchlight power removed.
START	START	Guarded, momentary-contact type switch used to operate the xenon arc lamp ignite circuit.
FOCUS	FOCUS	Momentary-contact type switch operates a focus-mechanism, nonreversible motor to change the light beam spread from 4° to 20°.
NOTE Light beam will continually cycle from large to small to large again as long as the FOCUS switch is held in the FOCUS position.		

SWITCH	POSITION	FUNCTION
Directional Control	DOWN	Operates the gimbal-mounted elevation motor to drive the searchlight max 70° down from level.
	UP	Operates the gimbal-mounted elevation motor to drive the searchlight max 10° up from level.
	LEFT-RIGHT	Operates the gimbal-mounted azimuth motor to drive the searchlight in the respective direction.
NOTE Mechanical stops limit searchlight travel in all directions. When a stop is reached the affected motor continues to rotate (as long as the switch is pressed) while the motor torque is absorbed by an integral friction clutch.		
IR ON/IR OFF	IR ON	IR filter closes over the lamp lens in approx 20 sec. The IR ON indicator light comes on.
	IR OFF	IR filter returns to the retracted position in approx 20 sec. The IR ON indicator light goes off.
BRIGHT/DIM	BRIGHT	Control panel illumination is bright.
	DIM	Control panel illumination is dim.

The four-way SEARCHLIGHT switch on the pilot's collective lever has the same function as the control panel directional control switch.

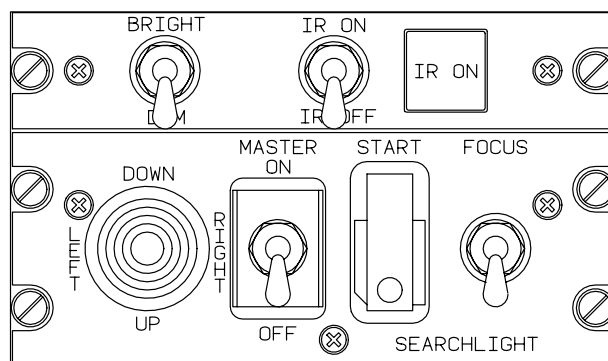
The searchlight lamp is supplied 28V DC power from the Starter Bus PP1 via a fuse in the Main Relay Box 1VE to the junction box and there via a circuit breaker and control circuit relay. The searchlight control circuit is supplied 28V DC power from the Auxiliary Bus PP22 via the SRCH LT circuit breaker on the center console switch panel to the control unit MASTER switch.

- NOTE**
- When the MASTER switch is ON, the optional cabin electrical heating (FMS 10-11) output is automatically halved.
 - Since the control circuit is powered by the Auxiliary Bus, the searchlight can only be operated when both generators or the ground EPU power the helicopter electrical system.

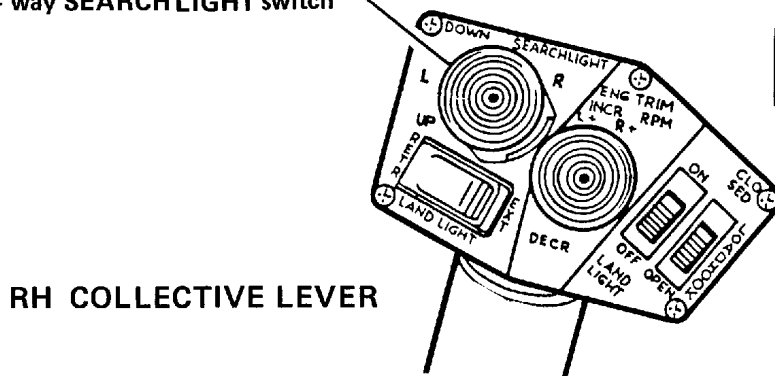


XENON SEARCHLIGHT

SEARCHLIGHT CONTROL PANEL



Four – way SEARCHLIGHT switch



RH COLLECTIVE LEVER

Fig. 1 SX-16 Searchlight - Center Controls